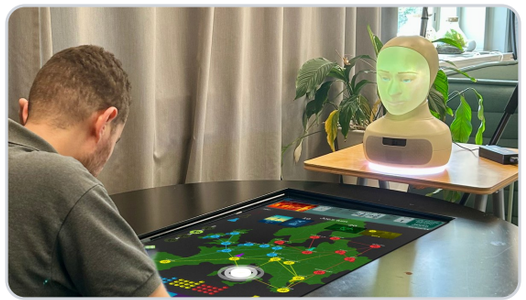
1 Testbed



state ↓ ↑ moves

rule-based solver

HRI corpus
70 sessions · 36 participants
speech · video · gaze · game logs

2 Partner signals

training time only

a Self-reported

enjoyment ●●●●○
rated once, after each game

speech · text → **Speech-to-Joy**

b Measured

brain activity
fMRI, only in the scanner

conversation → **brain encoder**

3 Model adaptation *current/future work*

trained

full-duplex h_t

partner audio
partner gaze
task state

LLM

text head
gaze head
action head

bias

robot speech
robot gaze
robot action

stage 1 · fine-tune
π · v self-play
gaze data
where · *abundant test-time inputs*

stage 2 · steer
recurrent intervention head
when · whether

80 ms

predictors